



EXPERIMENT - 2

Designing and Implementing ETL Processes Using Microsoft Power BI

Report



Subject Name: Business Intelligence & Tools
Subject Code: 24CSH-341

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1. Am

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

2. Objectives

- To understand the ETL process.
- To extract data from multiple worksheets.
- To perform data transformation using Power Query Editor.
- To create relationships among tables.
- To develop analytical measures using DAX.
- To prepare a dataset for reporting and decision-making.

3. Tools / Software Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor (built into Power BI Desktop)
- DAX (Data Analysis Expressions)

Dataset Source:

Global Superstore Dataset (Kaggle) —

<https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

4. Theory

ETL (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.

4.1 Extract Phase

The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.

4.2 Transform Phase

The transformation phase includes:

- Data cleaning and duplicate removal
- Handling missing values
- Data type correction
- Creating derived / calculated columns
- Filtering invalid records
- Data standardization
- Integrating multiple datasets

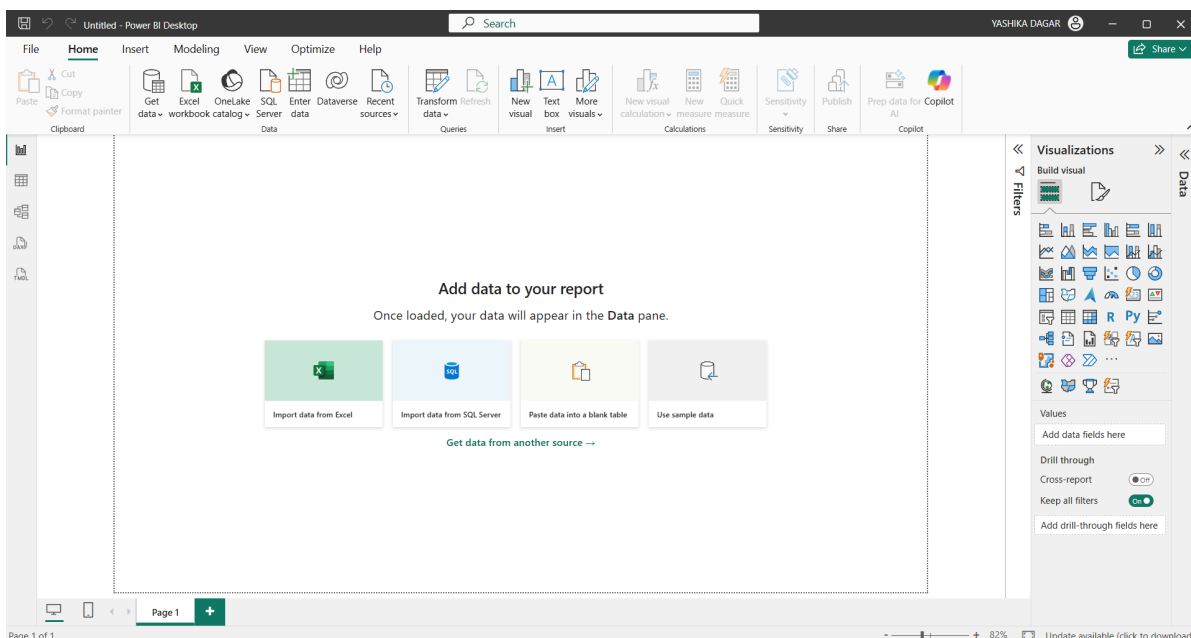
4.3 Load Phase

The transformed data is loaded into the Power BI data model, where relationships between tables can be established and analytical measures can be created using DAX. Power Query Editor serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

5. Step-wise Procedure

Step 1: Open Power BI Desktop

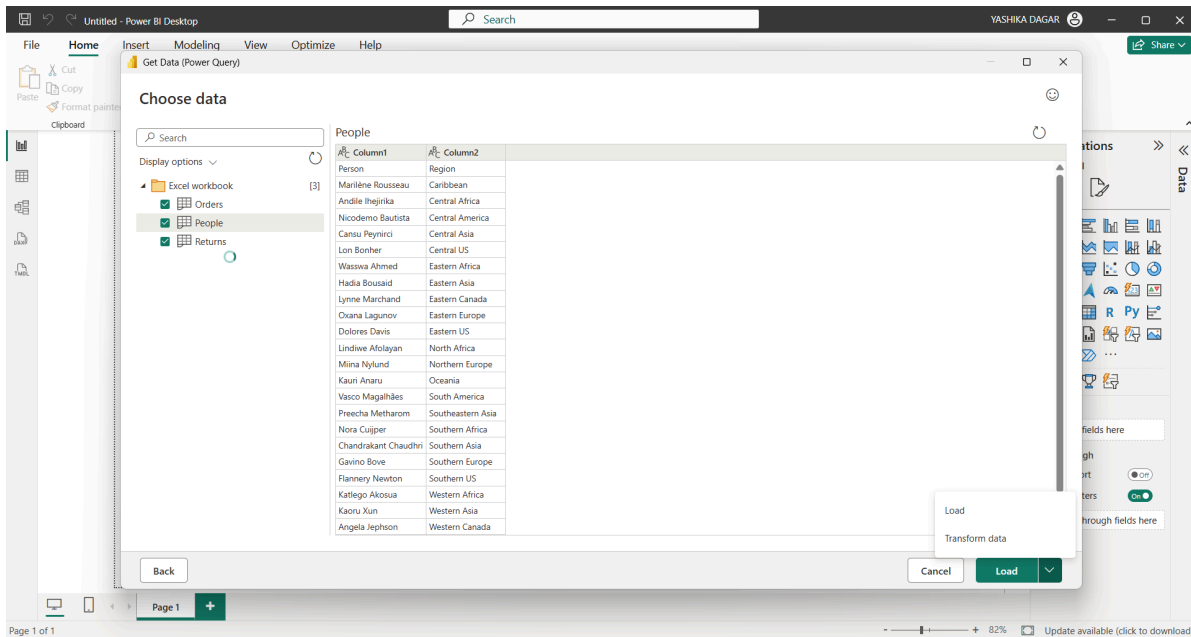
Launch Microsoft Power BI Desktop to begin the ETL workflow.



Step 2: Import the Dataset (Extract)

- Home → Get Data → Excel Workbook
- Select Global Superstore Dataset.xlsx

- Click Transform Data

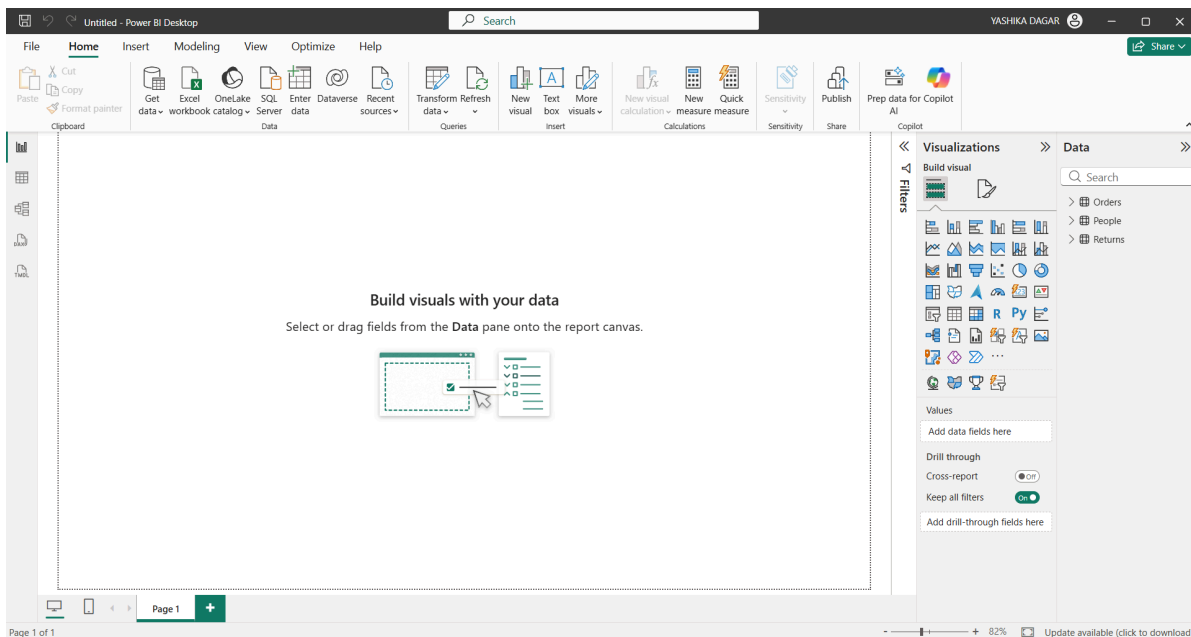


Step 3: Import Worksheets

The following worksheets were imported into Power BI:

- Orders
- Returns
- People

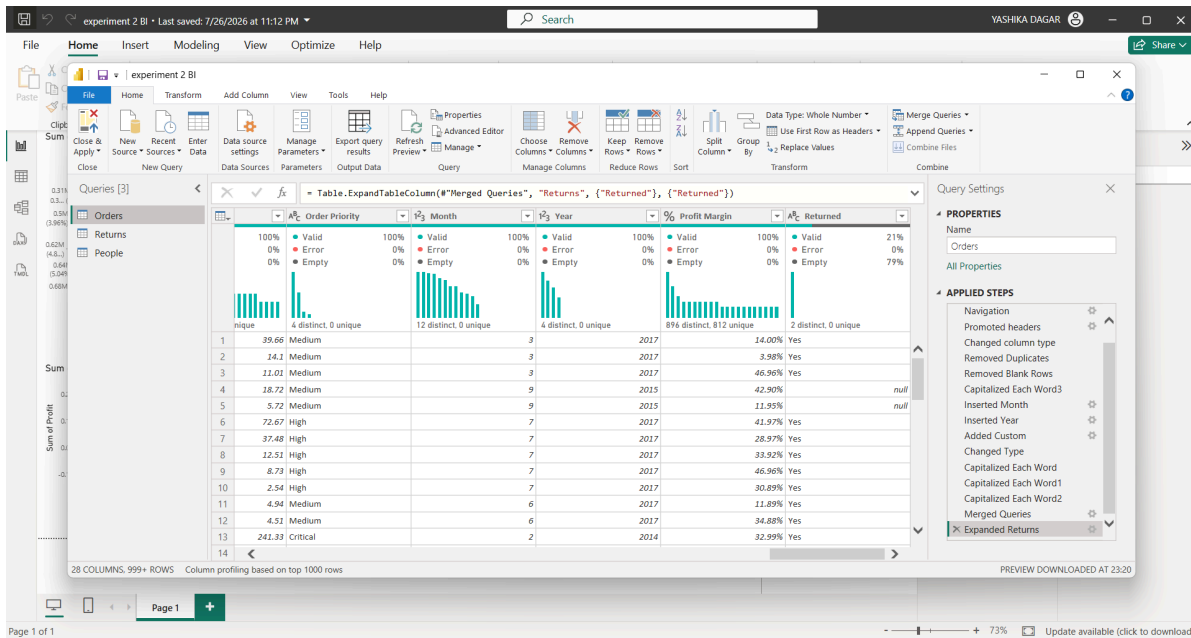
All imported tables were verified in the Queries pane.



Step 4: Transform the Orders Table

The following operations were performed on the Orders table using Power Query Editor:

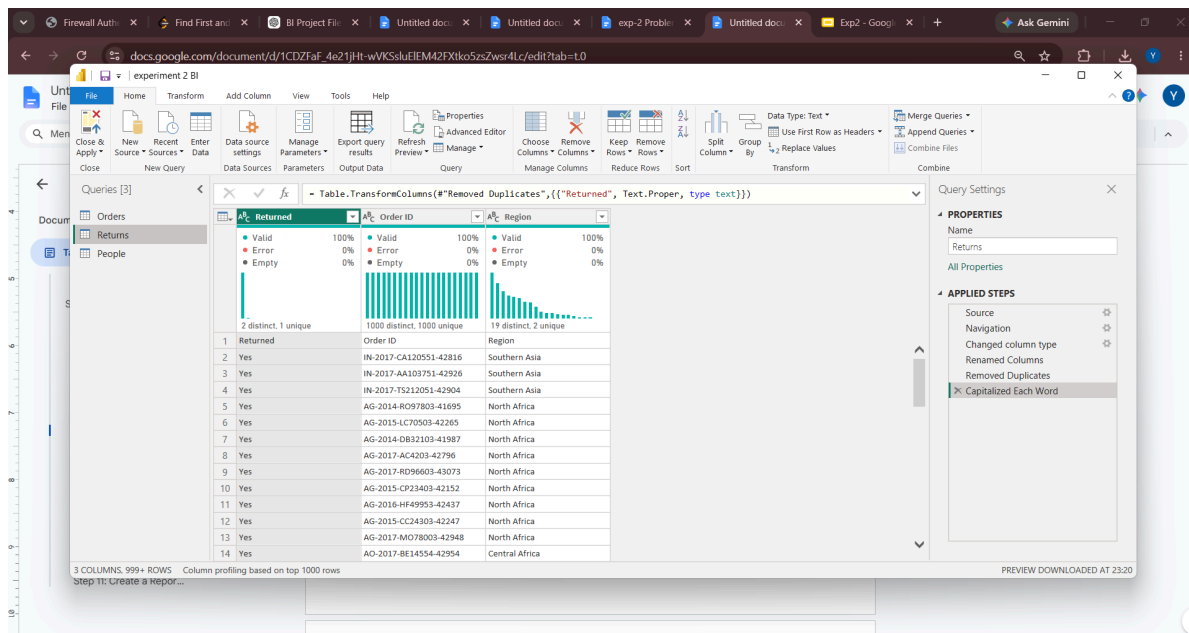
- Removed duplicate records
- Removed blank rows
- Verified and corrected data types
- Renamed columns where required
- Extracted Year and Month from Order Date
- Created a custom Profit Margin column
- Standardized Region and Segment values



Step 5: Transform the Returns Table

The following operations were performed on the Returns table:

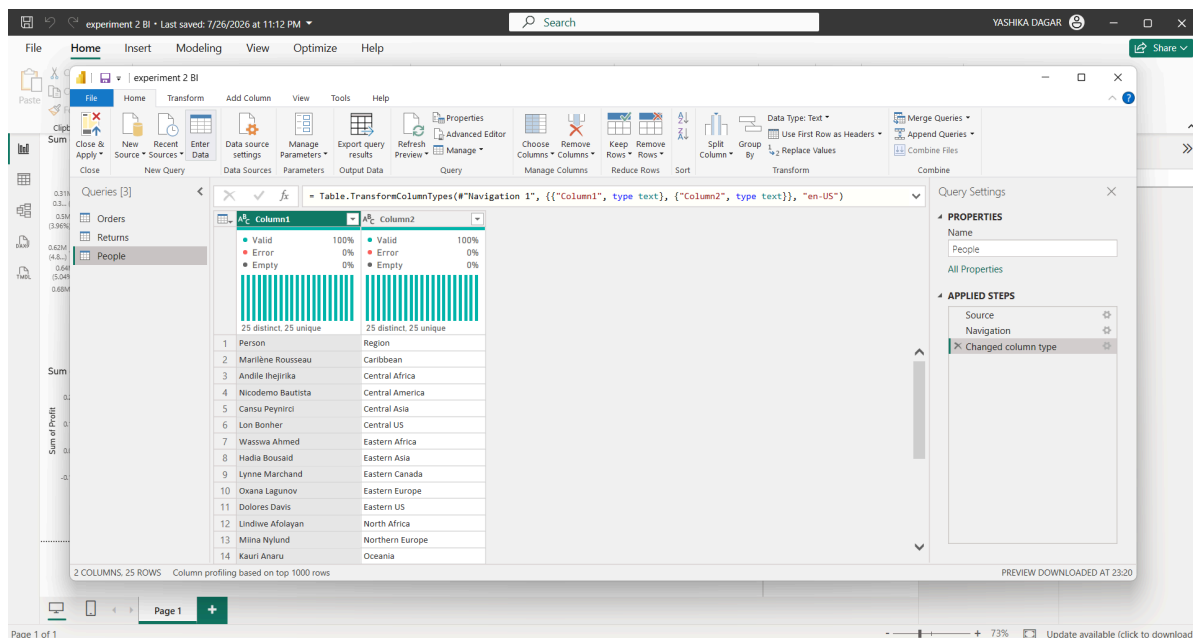
- Removed duplicate Order IDs
- Capitalized each word for consistency
- Renamed columns



Step 6: Transform the People Table

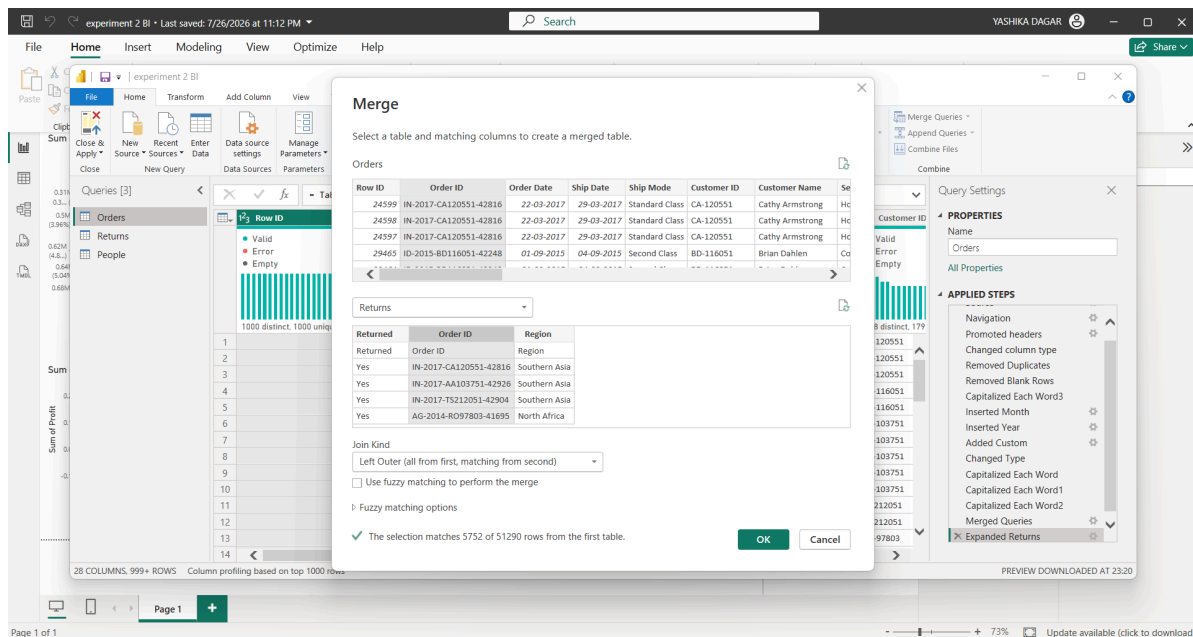
The People table was validated for regional consistency and the following operation was performed:

- Changed column data type



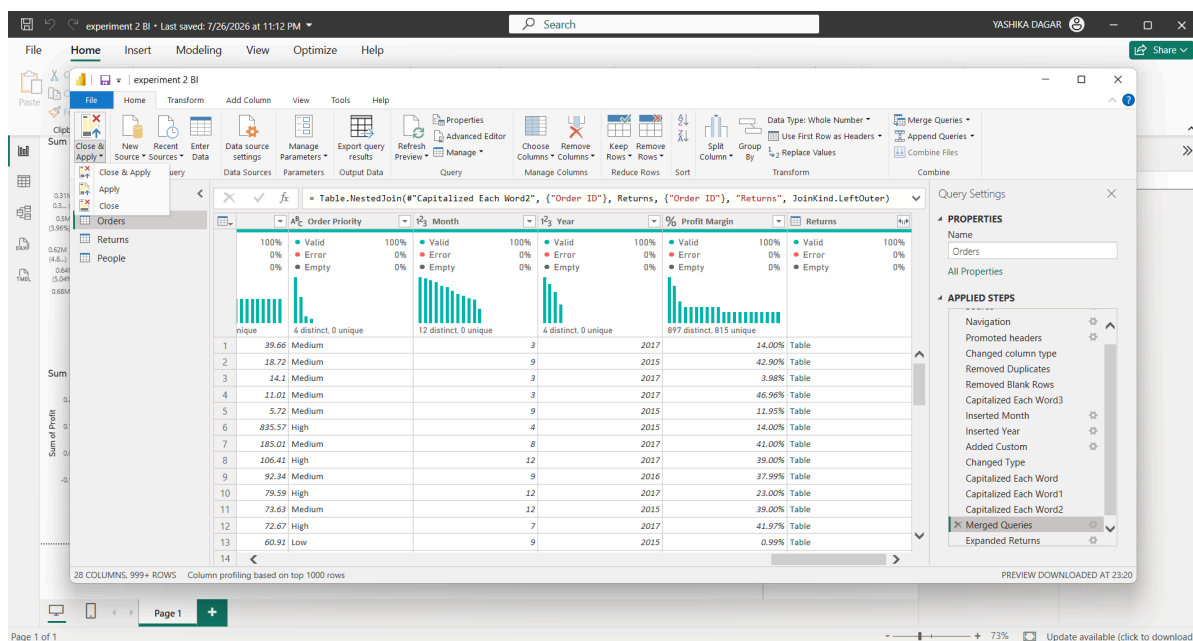
Step 7: Merge Tables

The Orders and Returns tables were merged using Order ID as the common key.



Step 8: Review Applied Steps

All transformation steps recorded in Power Query Editor were reviewed for accuracy, after which Close & Apply was clicked to load the transformed data into the Power BI data model.

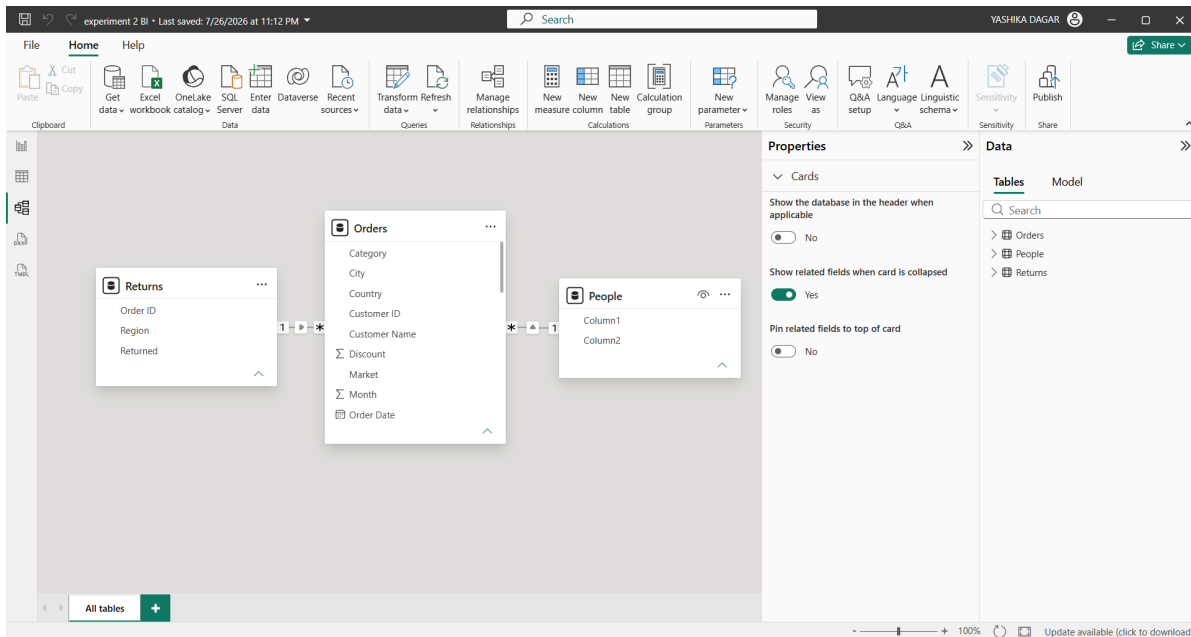


Step 9: Create Relationships (Load)

The following relationships were created in Model View to build a unified analytical data model:

- Orders ↔ Returns (Order ID)

- Orders ↔ People (Region)



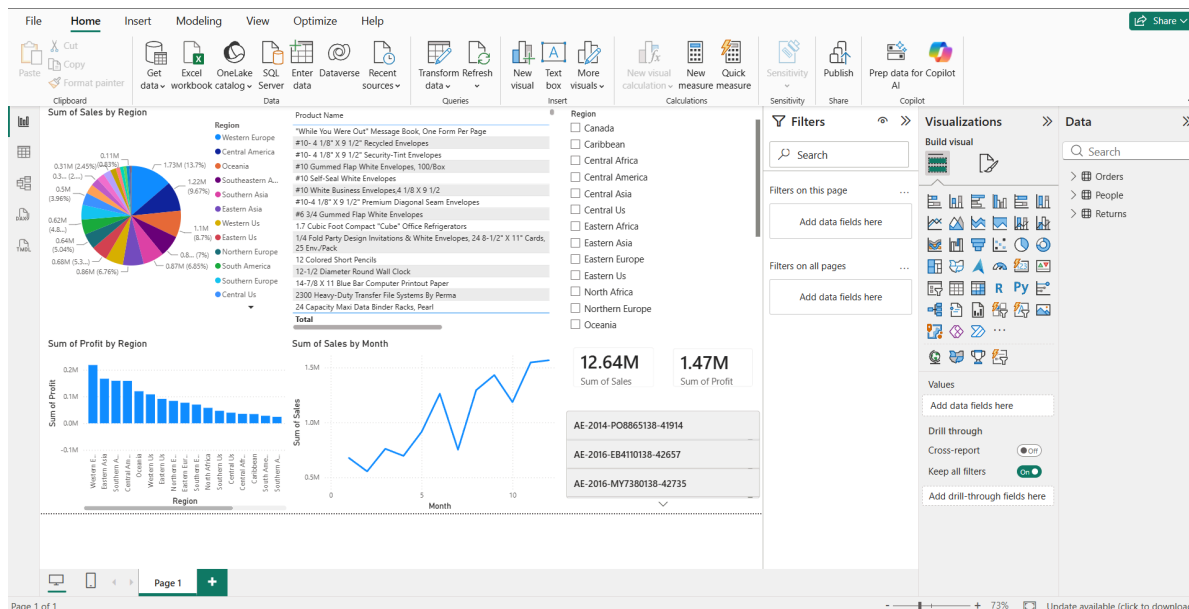
Step 10: Create DAX Measures

The following DAX measures were created to enable analysis and reporting:

```
Total Sales = SUM(Orders[Sales])
Total Profit = SUM(Orders[Profit])
Profit Margin = DIVIDE(SUM(Orders[Profit]), SUM(Orders[Sales]))
Total Orders = DISTINCTCOUNT(Orders[Order ID])
```

Step 11: Create a Report Visualization

A summary report was created using fields from the related tables, combining a pie chart of Sales by Region, a bar chart of Profit by Region, a line chart of Sales by Month, and KPI cards for Total Sales and Total Profit.



6. Result

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The resulting report visual shows the following headline figures for the dataset:

- Sum of Sales (overall): 12.64M
- Sum of Profit (overall): 1.47M
- Sales and Profit are broken down by Region (pie chart and bar chart) and by Month (line chart), enabling identification of high-performing and underperforming regions.

The final dataset was prepared and validated for reporting and business analysis.

7. Learning Outcomes

After completing this experiment, students will be able to:

- Explain the ETL process and its role in Business Intelligence systems. (CO1-BT2)
- Extract data from real-world business datasets using Power BI. (CO2-BT3)
- Apply transformation techniques including cleaning, filtering, and merging. (CO2-BT3)
- Create relationships and analytical measures in Power BI. (CO3-BT4)
- Validate transformed datasets for reporting and decision-making. (CO3-BT4)

8. Significance and Benefits of the Experiment

This experiment goes beyond a simple tool walkthrough — it builds practical, industry-relevant skills that are directly applicable to real-world data analytics roles.

8.1 Why This Experiment Matters

- ETL is the foundation of every BI and analytics system — almost no report or dashboard can be trusted without a properly cleaned and modeled dataset behind it.
- Power Query Editor is used identically across Power BI, Excel Power Pivot, and Power Automate, so the skills learned here transfer directly to other Microsoft data tools.
- Working with a multi-sheet, real-world dataset (Orders, Returns, People) mirrors how data actually arrives in organizations — messy, spread across sources, and in need of cleaning before analysis.
- Building relationships and DAX measures introduces the core logic used in almost all enterprise dashboards (KPIs, drill-downs, and cross-filtering).

8.2 Key Skills Gained

- Data cleaning: handling duplicates, blanks, and inconsistent text/number formats.
- Data transformation: creating calculated columns (e.g., Profit Margin, Year/Month extraction) that add analytical value.
- Data modeling: designing relationships (one-to-many) between fact and dimension tables.
- DAX fundamentals: writing aggregation, ratio, and distinct-count measures.
- Dashboard design: combining KPI cards, charts, and filters into one interactive report page.